

# Algonquin College

## School of Advanced Technology

Course: CST8238 – Web Programming

### Midterm Examination

Date: October 24th

Total Questions: 45 Multiple Choice Questions

Time Allowed: 90 Minutes

Instructor: Prof. Abul Qasim

---

### Instructions:

1. Answer all **45 questions**.
  2. Select the **one best answer** for each question (A, B, C, or D).
  3. Record your answers on the provided scantron or answer sheet.
  4. This exam covers:
    - HTML5 Semantics
    - CSS3 Styling & Layout Systems (Flexbox/Grid)
    - Bootstrap Framework
    - Basic JavaScript Logic
  5. Calculators and external devices are **not permitted**.
- 

### Part 1: HTML5 Structure & Semantics

1. Which declaration must appear at the very beginning of an HTML document to define it as HTML5?

- A. `<html version="5">`
- B. `<!DOCTYPE html>`
- C. `<head type="html5">`
- D. `<?xml version="1.0">`

2. Which element serves as the root container that wraps all visible and metadata content on a webpage?

- A. `<body>`
- B. `<head>`
- C. `<html>`
- D. `<root>`

3. Where should metadata, such as the character set and page title, be placed within an HTML document?

- A. Inside the `<body>` tag

- B. Inside the `<footer>` tag
- C. Inside the `<head>` tag
- D. Inside the `<nav>` tag

**4. Which semantic element is best suited for defining the introductory content of a page, typically containing the main heading and logo?**

- A. `<section>`
- B. `<header>`
- C. `<article>`
- D. `<aside>`

**5. To create a navigation menu, which semantic tag should wrap the list of links?**

- A. `<menu>`
- B. `<navigation>`
- C. `<nav>`
- D. `<dir>`

**6. Which attribute is used within an anchor `<a>` tag to create a link to a specific section on the same page?**

- A. `src="#section-id"`
- B. `link="#section-id"`
- C. `href="#section-id"`
- D. `ref="#section-id"`

**7. Which semantic element represents a self-contained composition, such as a blog post or news article, that could be distributed independently?**

- A. `<section>`
- B. `<div>`
- C. `<article>`
- D. `<span>`

**8. If you need to display content that is tangentially related to the main content, such as a sidebar or call-out box, which tag should be used?**

- A. `<sidebar>`
- B. `<aside>`
- C. `<section>`
- D. `<meta>`

**9. Which tag is used to define important text that should typically be rendered in bold?**

- A. `<b>`
- B. `<bold>`
- C. `<strong>`
- D. `<em>`

**10. Which element is used to define abbreviations or acronyms, allowing for a tooltip explanation?**

- A. `<acronym>`

- B. `<abbr>`
- C. `<short>`
- D. `<define>`

**11. In an HTML table, which tag defines a header cell that is typically bold and centered?**

- A. `<td>`
- B. `<tr>`
- C. `<th>`
- D. `<head>`

**12. Which list type is most appropriate for displaying a chronological work history?**

- A. Unordered List (`<ul>`)
- B. Definition List (`<dl>`)
- C. Ordered List (`<ol>`)
- D. Navigation List (`<nav>`)

**13. Which attribute is mandatory for the `<img>` tag to ensure accessibility for screen readers?**

- A. `title`
- B. `src`
- C. `alt`
- D. `description`

**14. To embed a video file with play and pause controls visible to the user, which attribute must be added to the `<video>` tag?**

- A. `play`
- B. `controls`
- C. `manage`
- D. `interface`

**15. Which form element is used to create a multi-line text input field?**

- A. `<input type="text">`
- B. `<textbox>`
- C. `<textarea>`
- D. `<input type="multiline">`

---

## Part 2: CSS Fundamentals & Styling

**16. How do you correctly link an external CSS file named `styles.css` to an HTML document?**

- A. `<style src="styles.css">`
- B. `<link rel="stylesheet" href="styles.css">`
- C. `<css link="styles.css">`
- D. `<script href="styles.css">`

**17. Which CSS selector is known as the "universal selector" and selects all elements on the page?**

- A. #all
- B. .reset
- C. \*
- D. body

**18. What is the correct syntax for selecting an element with the class name `container`?**

- A. #container
- B. container
- C. .container
- D. \*container

**19. Which CSS property is used to set the size of the text?**

- A. text-size
- B. font-size
- C. size
- D. font-weight

**20. To improve the readability of paragraph text, which property controls the vertical space between lines?**

- A. letter-spacing
- B. word-spacing
- C. line-height
- D. vertical-align

**21. Which box model property creates space *inside* an element's border?**

- A. margin
- B. border
- C. padding
- D. spacing

**22. Which box model property creates space *outside* an element's border?**

- A. margin
- B. border
- C. padding
- D. outline

**23. How do you horizontally center a block-level element with a fixed width?**

- A. margin: center;
- B. text-align: center;
- C. margin: 0 auto;
- D. padding: 0 auto;

**24. Which pseudo-class selector is used to define styles when a user hovers their mouse over an element?**

- A. :active
- B. :focus
- C. :hover

- D. :mouse

**25. Which property is used to remove bullet points from an unordered list?**

- A. list-style-type: none;
- B. bullet-style: none;
- C. list-type: empty;
- D. decoration: none;

**26. Which CSS property adds rounded corners to an element?**

- A. border-circle
- B. border-radius
- C. corner-round
- D. image-shape

**27. In table styling, which property removes the default space between adjacent cell borders?**

- A. border-spacing: 0;
  - B. border-collapse: collapse;
  - C. border-style: none;
  - D. table-layout: fixed;
- 

### **Part 3: Layout Systems (Flexbox & Grid)**

**28. Which CSS property value turns a container into a Flexbox container?**

- A. display: block;
- B. display: flex;
- C. layout: flex;
- D. position: flex;

**29. In a Flexbox container, which property aligns items along the main axis (e.g., horizontally in a row)?**

- A. align-items
- B. justify-content
- C. flex-direction
- D. flex-wrap

**30. Which Flexbox property allows items to wrap onto multiple lines if there isn't enough space?**

- A. flex-wrap: wrap;
- B. flex-flow: row;
- C. overflow: auto;
- D. white-space: normal;

**31. Which CSS property value turns a container into a Grid container?**

- A. display: grid;
- B. display: table;
- C. layout: grid;

- `D. position: grid;`

**32. Which CSS Grid property defines the number and size of columns in a grid?**

- A. `grid-columns`
- B. `grid-template-columns`
- C. `grid-layout-columns`
- D. `column-count`

**33. How do you make a grid item span across two columns?**

- A. `grid-column: 2;`
- B. `grid-span: 2;`
- C. `grid-column: span 2;`
- D. `col-span: 2;`

**34. Flexbox is generally best suited for which type of layout?**

- A. Two-dimensional layouts (rows and columns simultaneously)
- B. One-dimensional layouts (either a row or a column)
- C. Complex magazine-style layouts
- D. Database structure visualization

**35. CSS Grid is generally best suited for which type of layout?**

- A. Simple navigation bars
- B. One-dimensional lists
- C. Two-dimensional layouts (rows and columns simultaneously)
- D. Text typography only

**36. Which property adds consistent spacing between grid or flex items?**

- A. `spacing`
  - B. `gap`
  - C. `margin-between`
  - D. `padding-between`
- 

## **Part 4: Bootstrap & Responsive Design**

**37. Which meta tag is required in the `<head>` section to ensure proper rendering on mobile devices?**

- A. `<meta name="description">`
- B. `<meta name="viewport" content="width=device-width, initial-scale=1.0">`
- C. `<link rel="mobile" href="responsive">`
- D. `<script src="responsive.js">`

**38. How is the Bootstrap framework typically included in a project via CDN?**

- A. `<script src="bootstrap.js">`
- B. `<link href="bootstrap.css" rel="stylesheet">`
- C. `<import bootstrap>`
- D. `<style src="bootstrap">`

**39. Which Bootstrap class creates a fixed-width responsive container?**

- A. `.container-fluid`
- B. `.wrap`
- C. `.container`
- D. `.box`

**40. In the Bootstrap grid system, which class is used to create a horizontal group of columns?**

- A. `.grid`
- B. `.row`
- C. `.line`
- D. `.section`

**41. Which Bootstrap component is commonly used to create contained content boxes with headers and footers?**

- A. `.box`
- B. `.panel`
- C. `.card`
- D. `.frame`

**42. What is the standard conventional filename for the homepage of a website?**

- A. `home.html`
  - B. `main.html`
  - C. `index.html`
  - D. `default.html`
- 

## **Part 5: JavaScript Fundamentals**

**43. What is the output of the following JavaScript code snippet?**

```
let a = 10;  
let b = "20";  
console.log(a + b);
```

- A. 30
- B. "1020"
- C. NaN
- D. undefined

**44. Which of the following statements correctly describes the difference between `==` and `===` in JavaScript?**

- A. `==` checks only for value, while `===` checks both value and data type.
- B. `===` checks only for value, while `==` checks both value and data type.
- C. There is no difference; they are interchangeable.
- D. `==` is used for strings, while `===` is used for numbers.

**45. Which keyword is used to define a function in JavaScript?**

- A. `def`

- B. function
- C. func
- D. void

---

*End of Examination*